Snehal Jauhri

Post-Doctoral Researcher, 3D AI Lab

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Research Interests

• 3D Vision • Egocentric Perception • Generative AI • Agentic AI • Reinforcement Learning • Embodied AI

Experience

- **Technical University of Munich (TUM)** — *May 2026 - current*
  Post-Doctoral Researcher, 3D AI Lab (3dunderstanding.org) — Munich, Germany
  - 3D scene generation and understanding, world models, and 3D reinforcement learning
- **Allen Institute for AI (Ai2)** — *May 2025 - April 2026*
  PhD Research Scientist Intern, Perceptual Reasoning and Interaction Research (PRIOR) (prior.allenai.org) — Seattle, USA
  - Large-scale 3D simulation for visual sim-to-real robot learning
  - Reinforcement learning fine-tuning of Vision Language Action (VLA) models
- **Almende b.v.** — *2019 - 2021*
  Robotics Engineer — Rotterdam, Neterhlands
  - EU research projects on autonomous drone exploration (adacorsa.eu) and wheeled robots for homes
- **Bosch Engineering** — *2016 - 2018*
  Software Developer — Bangalore, India
  - GPS + IMU sensor fusion for localization of autonomous vehicles

Education

- **Technical University of Darmstadt** — *June 2021 - May 2026*
  PhD in Computer Science - AI for Robot Perception and Learning, PEARL Lab (pearl-lab.com) — Darmstadt, Germany
  - Grade: *Summa Cum Laude (mit Auszeichnung)*
  - Advisor: Prof. Georgia Chalvatzaki
  - Publications at ICCV, CoRL, RSS, ICRA, IROS, RA-L
- **Delft University of Technology** — *Aug 2018 - May 2020*
  Master of Science in Embedded Systems, Specialization: Robot Learning — Delft, Netherlands
  - Grade: 8.5/10 (Cum Laude)
- **National Institute of Technology Karnataka** — *2012 - 2016*
  Bachelor of Technology - Electrical & Electronics Engineering — Surathkal, India
  - Grade: 8.01/10 (First Class)

Research Activities

- **Presented PhD research at the Doctoral Consortium at ICCV 2025**
- **Workshops:**
  - Lead organizer of the 1st EgoAct Workshop on Egocentric Perception and Action for Robot Learning at RSS 2025 (egoact.github.io/rss2025)
  - Co-organizer of the 'Workshop on Mobile Manipulation and Embodied Intelligence (MOMA.v2): Integrating Perception, Learning, Control for Full Autonomy' at ICRA 2024 (mobile-manipulation.net/events/moma2024)
  - Co-organizer of the 'Workshop on effective Representations, Abstractions, and Priors for Robot Learning (RAP4Robots)' at ICRA 2023 (sites.google.com/view/rap4robots)
  - Co-organizer of the 'Workshop on Mobile Manipulation and Embodied Intelligence: Challenges and Opportunities' at IROS 2022 (mobile-manipulation.net/events/moma2022)
- **EU projects:** PhD lab representative for the EU Horizon Research Project: MANiBOT (manibot-project.eu)
- **Peer-Review:** CVPR, ICCV, ECCV, CoRL, RSS, ICRA, IROS, RA-L

PUBLICATIONS

WHOLE-MoMa: Whole-Body Mobile Manipulation using Offline Reinforcement Learning on Sub-optimal Controllers — Webpage, Paper
Snehal Jauhri, Vignesh Prasad, and Georgia Chalvatzaki
arXiv preprint, arXiv:2604.12509, 2026

MolmoB0T: Large-Scale Simulation Enables Zero-Shot Manipulation — Webpage, Paper, Code
Snehal Jauhri*, Abhay Deshpande*, Maya Guru*, Rose Hendrix*, Ainaz Eftekhar, Rohun Tripathi, Max Argus, Jordi Salvador, Haoquan Fang, Matthew Wallingford, Wilbert Pumacay, Yejin Kim, Quinn Pfeifer, Ying-Chun Lee, Piper Wolters, Omar Rayyan, Mingtong Zhang, Jiafei Duan, Karen Farley, Winson Han, Eli Vanderbilt, Dieter Fox, Ali Farhadi, Georgia Chalvatzaki, Dhruv Shah, and Ranjay Krishna
arXiv preprint, arXiv:2603.16861, 2026

MolmoSpaces: Large-Scale Open Ecosystem for Robot Manipulation and Navigation — Webpage, Paper, Code
Yejin Kim, Wilbert Pumacay, Omar Rayyan, Max Argus,, Winson Han, Eli VanderBilt, Jordi Salvador, Abhay Deshpande, Rose Hendrix, Snehal Jauhri, Shuo Liu, Nur Muhammad Mahi Shafiullah, Maya Guru, Arjun Guru, Ainaz Eftekhar, Karen Farley, Donovan Clay, Jiafei Duan, Piper Wolters, Alvaro Herrasti, Ying-Chun Lee, Georgia Chalvatzaki, Yuchen Cui, Ali Farhadi, Dieter Fox and Ranjay Krishna
Robotics: Science and Systems (RSS) 2026

UniFField: A Generalizable Unified Neural Feature Field for Visual, Semantic, and Spatial Uncertainties in Any Scene — Webpage, Paper
Snehal Jauhri*, Christian Maurer*, Sophie Lueth, and Georgia Chalvatzaki
IEEE International Conference on Robotics and Automation (ICRA) 2026,
(Also presented at the Open-World 3D Scene Understanding workshop at ICCV 2025)

2HandedAfforder: Learning Precise Actionable Bimanual Affordances from Human Videos — Webpage, Paper, Code
Snehal Jauhri*, Marvin Heidinger*, Vignesh Prasad, and Georgia Chalvatzaki
International Conference on Computer Vision (ICCV) 2025,
(Also presented at the EgoAct Workshop at RSS 2025)

6DOPE-GS: Online 6D Object Pose Estimation using Gaussian Splatting — Webpage, Paper
Yufeng Jin, Vignesh Prasad, Snehal Jauhri, Mathias Franzius, and Georgia Chalvatzaki
International Conference on Computer Vision (ICCV) 2025,
(Also presented at the RSS 2025 Gaussian Representations Workshop)

Learning Any-View 6DoF Robotic Grasping in Cluttered Scenes via Neural Surface Rendering — Webpage, Paper, Code
Snehal Jauhri, Ishikaa Lunawat, and Georgia Chalvatzaki
Robotics: Science and Systems (RSS) 2024,
(Also presented at the CVPR 2023 workshop on 3D Vision & Robotics)

Act*Per*MoMa: Active-Perceptive Motion Generation for Mobile Manipulation — Webpage, Paper, Code
Snehal Jauhri*, Sophie Lueth*, and Georgia Chalvatzaki
IEEE International Conference on Robotics and Automation (ICRA) 2024,
(Also presented at the RSS 2023 workshop on Taking Mobile Manipulators into the Real World)

Safe Reinforcement Learning of Dynamic High-Dimensional Robotic Tasks: Navigation, Manipulation, Interaction — Webpage, Paper, Code
Puze Liu, Kuo Zhang, Davide Tateo, Snehal Jauhri, Ziyuan Hu, Jan Peters, and Georgia Chalvatzaki
IEEE International Conference on Robotics and Automation (ICRA) 2023

Robot Learning of Mobile Manipulation with Reachability Behavior Priors — Webpage, Paper, Code
Snehal Jauhri, Jan Peters, and Georgia Chalvatzaki
IEEE Robotics and Automation Letters (RA-L) & IROS 2022
Best Mobile Manipulation Paper Award 🏆
(Also presented at the ICRA 2022 workshop on Behaviour Priors in Reinforcement Learning for Robotics)

Regularized Deep Signed Distance Fields for Reactive Motion Generation — Webpage, Paper, Code
Puze Liu, Kuo Zhang, Davide Tateo, Snehal Jauhri, Jan Peters, and Georgia Chalvatzaki
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022

Interactive Imitation Learning in State-Space — Paper, Code
Snehal Jauhri, Carlos Celemin, and Jens Kober
Conference on Robot Learning (CoRL) 2020,
(Also presented at the RSS 2020 workshop on Advances & Challenges in Imitation Learning for Robotics)

AWARDS AND HONORS

- **Doctoral Consortium** — *October 2025*
  International Conference on Computer Vision (ICCV) 2025
  - Selected to be part of the Doctoral Consortium at ICCV 2025 and present my PhD research titled: "Visual Robot Learning for Households: Coupling Perception, Mobility & Manipulation skills"

- **Best Paper Award** *October 2022*
  International Conference on Intelligent Robots and Systems (IROS) 2022
  - Best Mobile Manipulation paper award for "Robot Learning of Mobile Manipulation with Reachability Behavior Priors"
- **Best Paper Award (Finalist)** *September 2025*
  Human to Robot (H2R) workshop at the Conference on Robot Learning (CoRL) 2025
  - Best paper award finalist for "2HandedAfforder: Learning Precise Actionable Bimanual Affordances from Human Videos"
- **Best Project Award** *September 2023*
  ELLIS Summer School on Large-Scale AI for Research & Industry 2023
  - For the project "Camera Pose Refinement for Improved Radiance Fields"

Teaching/Mentoring

- **Teaching:**
  - **Statistical Machine Learning:** Lead Teaching Assistant, Lectures/Assignments on SVMs & PCA, 2022
  - **Robotic Perception and Manipulation:** Lead Teaching Assistant, Lectures on Robot Perception and Geometric Deep Learning, 2022 - 2024
- **Supervised Master Theses:**
  - **Dustin Gorecki** "Dexterous Bimanual Manipulation from Egocentric Human Demonstrations", 2026
  - **Christian Maurer** "Visual and Spatial Uncertainties in Any scene via Learned Feature Fields", 2025
  - **Marvin Heidinger** "Learning Precise Actionable Bimanual Affordances from Human Videos", 2025
  - **Sabin Grube Doiz** "Robot learning with egocentric 3D vision for mobile manipulation in homes", 2024
  - **Lars Pühler** "Open Vocabulary Navigation for Mobile Manipulation", 2024
  - **Maximilian Niessing** "Attention-based Object Pose Estimation in Cluttered Scenes", 2023
  - **Jan Schneider** "Model Predictive Policy Optimization Amidst Inaccurate Models", 2022

Technical Skills

- **Programming Languages:** Python, C
- **Libraries and Frameworks:** Pytorch, Open3D, stable-baselines, trimesh, OpenCV, nerfstudio, ROS
- **Simulators:** NVIDIA Isaac Sim, Isaac Gym, Mujoco, PyBullet
- **Developer Tools:** Docker, Git, Conda, SLURM, Claude Code, Agentic AI
- **Robots:** Tiago++ Mobile Manipulator, Franka Panda, KUKA LBR iiwa

References

1. **Prof. Angela Dai** (3dunderstanding.org/team.html)
   Associate Professor, 3D AI Lab
   Technical University of Munich (TUM)
   Email: angela.dai@tum.de
   Advisor: Post-Doc
2. **Prof. Georgia Chalvatzaki** (pearl-lab.com/people/georgia-chalvatzaki)
   Full Professor, Interactive Robot Perception & Learning (PEARL) Lab
   Technical University of Darmstadt (TU Darmstadt)
   Email: georgia.chalvatzaki@tu-darmstadt.de
   Supervisor: PhD Thesis
3. **Dr. Rose Hendrix** (rosehendrix.com)
   Research Scientist, Perceptual Reasoning and Interaction Research (PRIOR)
   Allen Institute for AI (Ai2)
   Email: roseh@allenai.org
   Supervisor: PhD Research Internship
4. **Prof. Jens Kober** (jenskober.de)
   Associate Professor, Cognitive Robotics department (CoR)
   Delft University of Technology (TU Delft)
   Email: j.kober@tudelft.nl
   Supervisor: Master Thesis